DILI — Classic Offenders (Drug Hall)

1. DILI Drug Hall banner

WHAT YOU SEE

DILI DRUG HALL — High-Yield Drugs for Boards sign

WHAT IT ENCODES

DILI is a diagnosis of exclusion — before blaming any drug in this hall, exclude viral hepatitis (A/B/C/E), autoimmune hepatitis (ANA/ASMA), and biliary obstruction (RUQ ultrasound). Then classify by R-value pattern and assess severity with Hy's Law (hepatocellular ALT >3x ULN + bilirubin >2x ULN = >10% mortality).

BOARD TRAP

BOARD TRAP: with a single suggestive drug on the med list, the wrong move is to stop the workup; you must still document a temporal relationship and rule out competing causes, since untreated viral or obstructive disease is the real miss.

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2. Room 1: INH (isoniazid)

WHAT YOU SEE

Isoniazid figure, ALT up to 6 months latency

WHAT IT ENCODES

Isoniazid causes a HEPATOCELLULAR (R >5) idiosyncratic-metabolic injury via a toxic acetylhydrazine metabolite. About 10–20% develop transient asymptomatic ALT elevation (adaptation), but ~0.5–1% develop serious hepatitis; risk climbs with age (>35), daily alcohol, and combination with rifampin/pyrazinamide.

BOARD TRAP

WRONG to stop INH for an isolated asymptomatic ALT <3x ULN — that is adaptation and you continue with monitoring; the discriminator for stopping is ALT >3x ULN WITH symptoms/jaundice or >5x ULN regardless of symptoms.

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3. INH true hepatitis criteria

WHAT YOU SEE

TRUE HEPATITIS: latency weeks-6mo, jaundice, ALT >3x ULN

WHAT IT ENCODES

True INH hepatitis has a latency of weeks to 6 months and presents with anorexia, nausea, jaundice, and ALT >3x ULN. Stop INH immediately for symptomatic hepatitis, jaundice, ALT >3x ULN with symptoms, or >5x ULN asymptomatic; combined with rifampin and pyrazinamide the hepatotoxicity is additive.

BOARD TRAP

WRONG to expect an allergic picture — INH injury is metabolic, so there is NO rash, fever, or eosinophilia; their absence does not make it benign, and waiting for allergic signs delays appropriate discontinuation.

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4. Room 2: Nitrofurantoin

WHAT YOU SEE

Nitrofurantoin bottle with wolf, AUTOIMMUNE HEPATITIS MIMIC

WHAT IT ENCODES

Nitrofurantoin (long-term UTI prophylaxis, classically older women) causes CHRONIC hepatitis that mimics autoimmune hepatitis with positive ANA/ASMA, interface hepatitis, and plasma cells. It also causes interstitial pulmonary fibrosis — a signature dual toxicity.

BOARD TRAP

WRONG to diagnose primary AIH and start indefinite immunosuppression in a woman on chronic nitrofurantoin — stop the drug first; drug-induced AIH resolves on withdrawal and rarely relapses off steroids, unlike idiopathic AIH.

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5. Nitrofurantoin lung companion

WHAT YOU SEE

Lungs labeled PULMONARY FIBROSIS

WHAT IT ENCODES

Nitrofurantoin's other classic organ target is the lung: acute hypersensitivity pneumonitis (eosinophilia, fever) with short-term use and chronic interstitial pulmonary fibrosis with long-term prophylaxis. Pair 'chronic hepatitis + pulmonary fibrosis' as the nitrofurantoin combination.

BOARD TRAP

WRONG to attribute new dyspnea and a restrictive PFT pattern in a chronic-nitrofurantoin patient to age or heart failure — drug-induced pulmonary fibrosis is the board answer, and it can coexist with the hepatic AIH-like injury.

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6. Room 3: Amox-clav (#1 idiosyncratic)

WHAT YOU SEE

Pickle knight AMOX-CLAV, DELAYED damage after stopping

WHAT IT ENCODES

Amoxicillin-clavulanate is the #1 cause of idiosyncratic DILI in the West. It produces a CHOLESTATIC or MIXED pattern (ALP/GGT up, R <2 to 2–5) with jaundice and pruritus, and onset is characteristically DELAYED — days to weeks AFTER the antibiotic course is finished. The clavulanate component drives the injury (HLA-linked).

BOARD TRAP

WRONG to exclude amox-clav just because the patient already completed the course — delayed onset after stopping is its signature; failing to connect a recent antibiotic course to new cholestatic jaundice is the classic miss.

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7. Vanishing bile duct syndrome

WHAT YOU SEE

Skull + VANISHING BILE DUCT SYNDROME

WHAT IT ENCODES

Severe cholestatic DILI (amox-clav, and others like flucloxacillin, carbamazepine, azithromycin) can cause progressive bile-duct loss = ductopenia / vanishing bile duct syndrome, with prolonged cholestasis that may progress to biliary cirrhosis.

BOARD TRAP

WRONG to expect rapid normalization in every cholestatic DILI — vanishing bile duct syndrome can cause cholestasis persisting for months after the drug is stopped, and a slow recovery does not mean the diagnosis is wrong.

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8. Room 4: Statins

WHAT YOU SEE

Statin knight, LIPID LOWERING CHAMPION, mild ALT up

WHAT IT ENCODES

Statins cause a mild, usually asymptomatic ALT rise (often <3x ULN) that frequently represents adaptation; clinically serious DILI is rare (~1 in 100,000). Routine periodic LFT monitoring is NO longer recommended — check baseline ALT, then only if symptoms arise.

BOARD TRAP

WRONG to stop a statin in a stable patient with mild ALT elevation <3x ULN — this is the most-tested statin trap; you continue and monitor, since cardiovascular benefit far outweighs the rare hepatotoxicity.

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9. Statin liver disease note

WHAT YOU SEE

LIVER DISEASE CONTRAINDICATION crossed; stable chronic liver disease NOT a contraindication

WHAT IT ENCODES

Stable/compensated chronic liver disease — including NAFLD/NASH, chronic HCV, and compensated cirrhosis — is NOT a contraindication to statins and statins may even benefit NAFLD. Avoid statins only in decompensated cirrhosis or acute liver failure.

BOARD TRAP

WRONG to withhold a statin from a patient with compensated NAFLD or stable cirrhosis who needs it for ASCVD risk — denying indicated statin therapy in compensated CLD is the wrong board answer.

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10. Room 5: Valproate (VPA)

WHAT YOU SEE

Valproate balloon causes STEATOSIS, MITOCHONDRIAL TOXICITY

WHAT IT ENCODES

Valproic acid causes MICROVESICULAR steatosis through mitochondrial beta-oxidation toxicity; it can cause severe, even fatal, hepatotoxicity and hyperammonemic encephalopathy (treat with L-carnitine). Hepatotoxicity is typically NOT predicted by routine LFTs and may be idiosyncratic.

BOARD TRAP

WRONG to reassure based on a normal ammonia or mildly abnormal LFTs in a confused patient on valproate — valproate-induced hyperammonemic encephalopathy can occur with normal LFTs, and the treatment is L-carnitine plus drug withdrawal, not just monitoring.

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11. VPA high-risk groups

WHAT YOU SEE

HIGH-RISK GROUPS: children <2 yrs, polytherapy, mitochondrial/metabolic disorders

WHAT IT ENCODES

Fatal valproate hepatotoxicity risk is highest in children <2 years old, those on anticonvulsant polytherapy, and patients with inherited mitochondrial disease (notably POLG/Alpers gene mutations), in whom valproate is contraindicated.

BOARD TRAP

WRONG to start valproate in a young child with developmental regression/seizures of unknown cause without considering POLG mutation — valproate can precipitate fatal liver failure (Alpers syndrome) and is contraindicated in known mitochondrial disease.

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12. Room 6: Amiodarone

WHAT YOU SEE

Amiodarone crab, causes STEATOSIS (macrovesicular)

WHAT IT ENCODES

Amiodarone causes MACROVESICULAR steatosis/steatohepatitis with phospholipidosis, lamellar inclusion bodies, and Mallory bodies — histology that closely mimics alcoholic hepatitis. It is highly lipophilic, accumulates in tissue, and has a half-life of ~weeks to months.

BOARD TRAP

WRONG to diagnose alcoholic hepatitis in a non-drinker on amiodarone with steatohepatitis and Mallory bodies — only amiodarone shows the phospholipid LAMELLAR inclusion bodies on EM; the drug history and lamellar bodies are the discriminators.

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13. Amiodarone NOT immune

WHAT YOU SEE

NOT IMMUNE; can cause cirrhosis even after stopping

WHAT IT ENCODES

Amiodarone toxicity is metabolic/accumulation-related, NOT immune, so there is no fever/rash/eosinophilia. Because of tissue accumulation and a long half-life, hepatic (and pulmonary and thyroid) toxicity can progress to cirrhosis and continue even AFTER the drug is discontinued.

BOARD TRAP

WRONG to exclude amiodarone as the cause because it was stopped weeks-to-months ago — it persists in tissue and injury can appear or worsen long after discontinuation, unlike most drugs that improve on dechallenge.

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14. R-value key (pattern reminders)

WHAT YOU SEE

Bottom key: hepatocellular ALT/ALT up, cholestatic ALP/GGT up, R-value formula

WHAT IT ENCODES

R-value = (ALT ÷ ULN) ÷ (ALP ÷ ULN). R >5 = hepatocellular (ALT-dominant, e.g., INH, statins, APAP), R 2–5 = mixed, R <2 = cholestatic (ALP/GGT-dominant, e.g., anabolic steroids, estrogens; amox-clav is cholestatic/mixed). The R-value at presentation defines the pattern.

BOARD TRAP

WRONG to compute R from AST instead of ALT, or to skip GGT confirmation — R uses ALT, and an isolated ALP rise needs a high GGT to confirm a hepatic (rather than bony) source before being called cholestatic.

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15. STOP sign + monitor

WHAT YOU SEE

Red STOP sign: stop drug, withdraw if enzymes >5x ULN

WHAT IT ENCODES

Management of all DILI is to STOP the offending drug first, then support. Withdraw the drug if transaminases reach ≥5x ULN asymptomatic, or >3x ULN with symptoms/jaundice (Hy's Law concern); NAC is specific only for acetaminophen, and steroids only for hypersensitivity/AIH-like/checkpoint-inhibitor injury.

BOARD TRAP

WRONG answer: continuing the drug while merely 'monitoring' once thresholds are crossed or jaundice appears — at ≥5x ULN, or >3x ULN with symptoms/jaundice, prompt discontinuation is required, not watchful waiting.